

‘Forces make objects move’—this is a pre-Newtonian idea. Modern mechanics began when Newton developed a **complete ideology** for discussing motion, one that we still study today **because it continues to allow us to make accurate predictions**, e.g. ‘this bullet will travel this far into a sandbank when fired from this distance away at this velocity’. This complete ideology begins with ...

NEWTON’S FIRST LAW: An object either remains at rest or continues to move at a constant velocity, unless acted upon by a **force**.

So, forces make objects **accelerate**. Or, rather, an object travel at constant velocity until a **net** or **resultant** force acts upon it.

Whether or not forces ‘exist’ in real life is a philosophical discussion possibly worth having, but the point is we don’t **need** to, in order to ‘do the maths’. Whatever your stance is, it doesn’t stop you from accepting forces as part of the **Newtonian Model**, and carrying out calculations using ...

NEWTON’S SECOND LAW: The acceleration given to an object acted on by one or more forces is given by the relationship

$$\text{Resultant force} = \text{mass of the object} \times \text{acceleration}$$

But be clear in how you think about this: the game (namely, using the Newtonian Model) absolutely requires us to think of forces causing acceleration, not the other way round. This seems rather weird: we can more or less measure acceleration directly in experiment, but not forces, so it is not as though we can ‘see’ a force of 10 N acting on an object and then predict it will have a certain acceleration. All we can do is verify that a Newtonian analysis is consistent and useful.

To use the model correctly, you have to accept a few rules about how forces ‘work’ in this game.

To analyse an object’s motion, only forces **external** to the object are relevant.

Think about a child sitting in a supermarket trolley, pushing forward on the handlebars.

Any force is said to be exerted **by** some object A **on** an object B.

For example: ‘the force exerted **by** my hand **on** the bicycle saddle’. In our game, you should be able to apply this analysis to any force which is part of the model. This includes forces with abbreviated descriptions, e.g. ‘weight’ or ‘driving force’ or ‘friction’.

Almost all forces exerted by an object A on object B require A to **be in direct contact with** B. The main exception is an object’s **weight**.

So virtually all forces are contact forces. On the other hand, my piano’s weight can be thought of as the force exerted **by** the Earth **on** the piano (from afar).

Near the surface of the earth, the weight of an object is roughly 9.8 Newtons per kilogram of mass.

An important empirical rule. This was discovered when it was observed that all objects in freefall accelerate downwards at the same rate.